NP VISMAYA



MRITI YANTRA

Growing Naturally for a Sustainable Future

THE PROBLEM

01. UNPREDICTABLE CLIMATE
CHANGING HUMIDITY, FLUCTUATING TEMPERATURES, IRREGULAR STORMS

02. PRESSURE ON LAND
URBANISATION, INDUSTRIALISATION

03. NO FIELD AWARENESS
WEATHER FORECASTS TELL YOU WHAT HAPPENS ACROsS A REGION BUT A PARTICULAR FIELD CAN HAVE DIFFERENT CONDITIONS





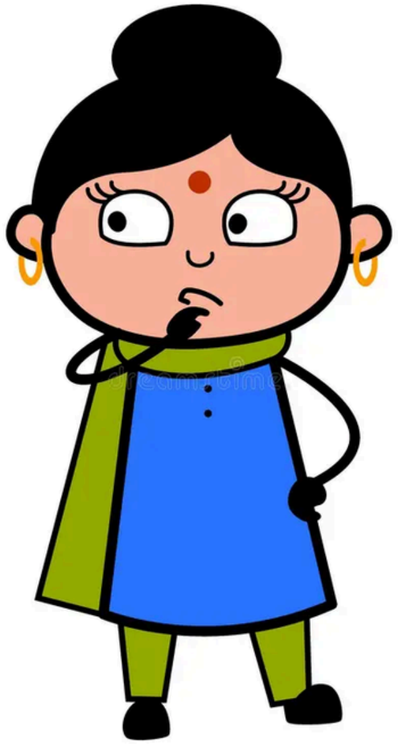
THE PROBLEM

04. SMALLHOLDER FARMERS
AGRICULTURAL TECHNOLOGY IS EXPENSIVE, FRAGMENTED, DIFFICULT TO ACCESS, DIFFICULT IN SMALL SCLAES

05. COMBINED CONDITIONS
SOIL,SUNLIGHT, TEMPERATURE, HUMIDITY ALL HAVE A COMBINED EFFECT

06. CHANGING SOIL
SOIL QUALITY CHANGES WITH SEASONS AND REQUIRES RESPECTIVE ANALYSIS

WHAT'S WRONG WITH EXISTING SOLUTIONS?



01. NOT FIELD SPECIFIC

REGIONAL WEATHER DATA MAY NOT REFLECT ACTUAL CONDITIONS OF THE FEILD

02. EXPENSIVE

ADVANCED PRECISION AGRICULTURE SYSTEMS CAN BE TOO COSTLY FOR SMALLHOLDER FARMERS

03. FRAGMENTEED DATA

SOIL, WEATHER, ENVIRONMENTAL INFORMATION OFTEN COME FROM SEPARATE SOURCES RATHER THAN ONE INTEGRATED SYSTEM

04. DATA WITHOUT DECISIONS

MANY SYSTEMS PRODUCE MEASUREMENTS BUT NOT SIMPLE, ACTIONABLE RECOMMENDATIONS TO FARMERS

SOLUTION

MRITI YANTRA

MRITI YANTRA IS A LOW COST, MODULAR FIELD MONITORING SYSTEM THAT COMBINES MULTIPLE ENVIRONMENTAL PARAMETERS AND CONVERTS THEM INTO USEFUL INFORMATION AND AUTOMATED ACTION.

SENSE → PROCESS → INTERPRET → ACT

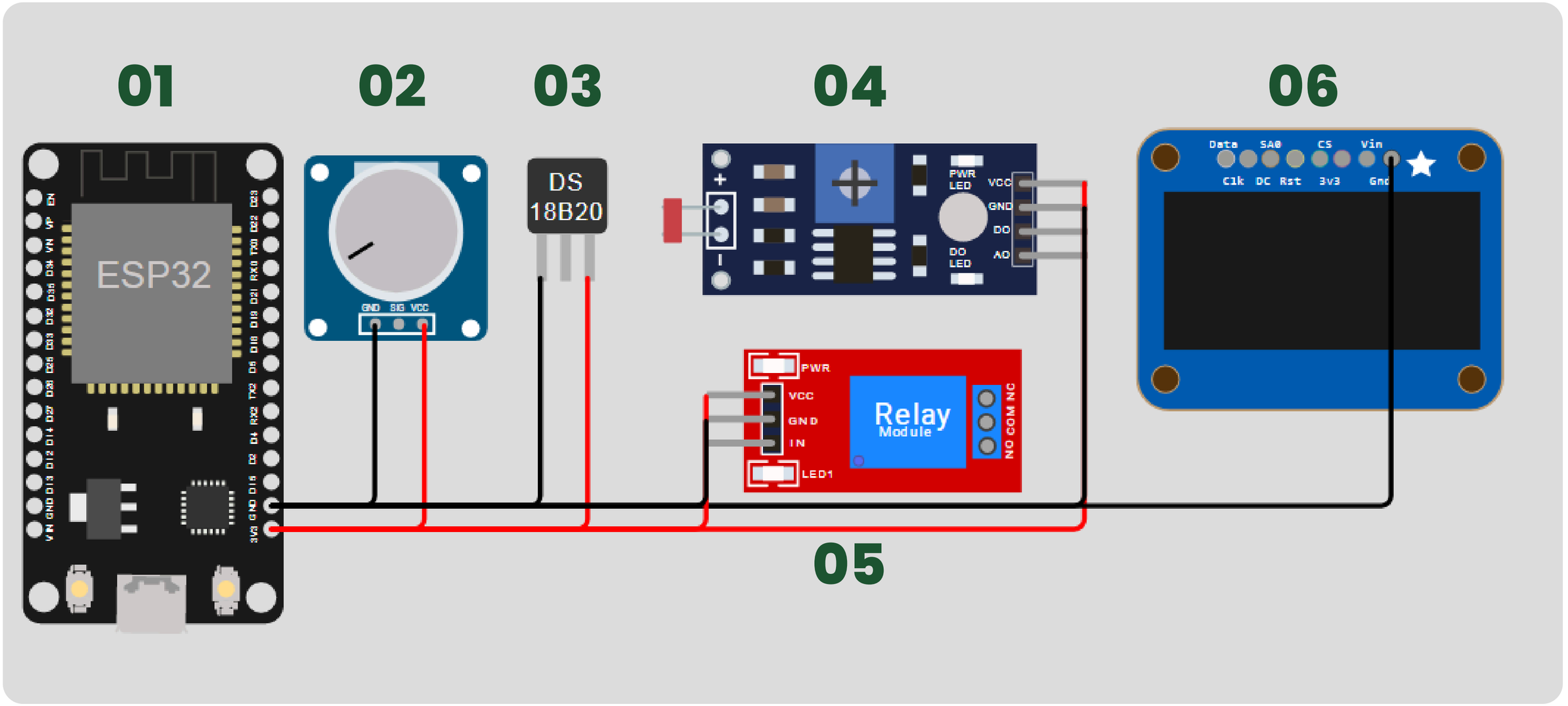
SENSE - IT SENSES SOIL CONDITIONS, LIGHT, TEMPERATURE, HUMIDITY AND OTHER ENVIRONMENTAL FACTORS.

PROCESS - THE ESP32, SENSORS AND RELAY MODULE HELP PROCESS THE COLLECTED INFORMATION.

INTERPRET - THE SYSTEM EVALUATES THE COMBINED CONDITIONS. THE COLLECTED DATE IS THEREIN PROCESSED WITH ADDITIONAL AI ASSISSTANCE.

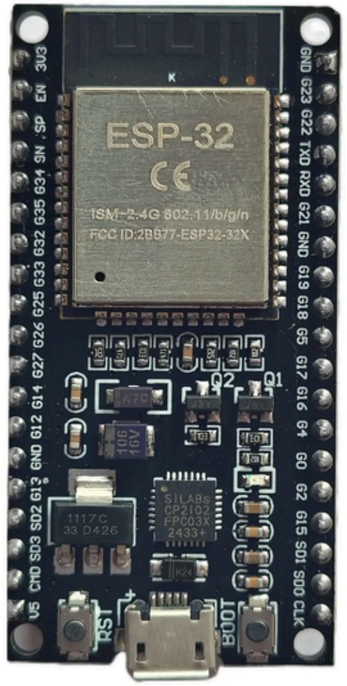
ACT - OLED PROVIDES FEEDBACK AFTER PROCESSING THE DATA AND THE REALY ENABLES AUTOMATED RESPONSE

COMPONENTS



*THE CIRCUIT WAS CREATED WITH THE HELP OF WOWKI.COM DUE TO SUBSTANTIAL COSTS INVOLVED.

*THIS IS A VERY BASIC PROTOTYPE AS SOFTWARE DOES NOT SUPPORT ADDITIONAL ADD ONS LIKE CAMERA AND KAGGLE DATABASE.



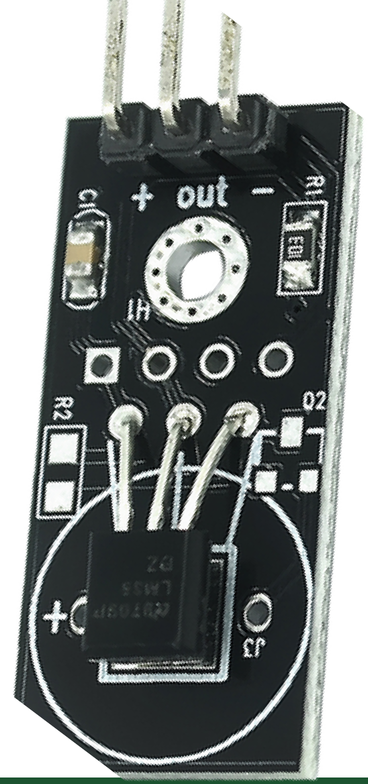
1. ESP32

- ESP32 IS THE MAIN MICROCONTROLLER.
- IT READS ALL THE SENSORS AND PROCESSES THE READINGS.
- IT CONTROLS THE RELAY AND SENDS INFORMATION TO OLED.
- IT HAS BUILT IN WIFI FOR OUR FUTURE WEATHER + AI FEATURES



2. POTENTIOMETER

- USING THIS WE SIMULATE THE ANALOG OUT PUT OF SOIL MOISTURE.
- TURNING THE KNOB CHANGES ANALOG VOLTAGE THUS SIMULATE DIFFERENT SOILD CONDITIONS.



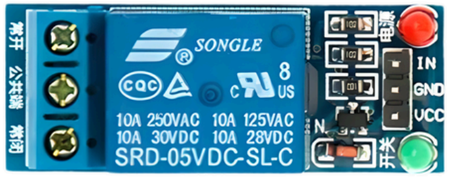
3.DS18B20

- THIS MEASURES THE TEMPERATURE.
- IT GIVES REAL TIME ENVIRONMENTAL/ SOIL - TEMPERATURE INFORMATION.



4. LDR/PHOTORESISTOR

- THIS HELPS MEASURE THE LIGHT INTENSITY.
- IT HELPS UNDERSTAND HOW MUCH SUNLIGHT ON AN AVERAGE IS RECEIVED BY THE SOIL.I .



5.RELAY MODULE

- IT ACTS AS A SWITCH CONTROLLED BY THE ESP32.
- IT WOULD HELP CONTROL ADDITIONAL ADD ONS SUCH AS WATER PUMP OR ANY OTHER AGRICULTURAL DEVICE.



6.OLED

- IT DISPLAYS THE READINGS OF THE SENSORS AND THE WEATHER.
- ALL SYSTEM INFORMATION IS PROJECTED HERE IN.

FUTURE VISION

TEH CURRENT PROTOYPE IS A VERY BASIC ONE AS CERTAIN FEATURES COULDNOT BE SIMULATED WITH THE AVAILABLE TECHNOLOGY AND THE ACTUAL PARTS COST A LOT MORE.

THE ACTUAL VISION IS TO ADD APPROPRIATE PHYSICAL SOIL SENSORS THAT SENSE PH MOISTURE, HUMIDITY,ETC. ALSO BASED ON SUN LIGHT REFLECTION DETERMINE THE LOOK OF THE SOIL AND COMPARE WITH OUR DATABASE. FURTHER WE D LIKE TO IMPLEMENT CONNCTED INTELLIGENCE TO GIVE LIVE WEATHER, FIELD SPECIFIC DATA AND AI ASSISSTED GUIDANCE IN REGIONAL LANGAUGE THROUGH PAGERS. ALSO UPDATE AND GIVE WARNINGS REGARDING ANY NATURAL DISASTER OR SUDDEN ONSETS OF WEATHER. THIS AIMS TO BUILID A SOLUTION WHERE FARMERS IN RURAL AREAS DONT FEAR TECHNOLOGY DUE TO ASSOCIATED COMPLEXITIES RATHER EMBRACE IT AND BUILD A HAPPY ECOSYSTEM!


THANK
YOU